

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456565.09032 +/- 0.00077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.049
Transit depth (Rp/Rs)^2	2.35 +/- 1.52 [%]
Orbital Inclination (inc)	86.43 +/- 1.7
Airmass coefficient 1 (a1)	0.82 +/- 0.55
Airmass coefficient 2 (a2)	0.16 +/- 0.81
Scatter in the residuals of the lightcurve fit is	69.45 %
Best Comparison Star	#2 - [513, 82]
Optimal Aperture	2.76
Optimal Annulus	6.75
Transit Duration (day)	0.008 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.049 vs 0.131 (deviation 0.41σ)
Duration fit vs published	0.008 d vs 0.1075 d (deviation 11.97σ)
Mid-time O–C	-5359.6 min
Depth SNR	2.8
Residual scatter	69.45 %

Light curve

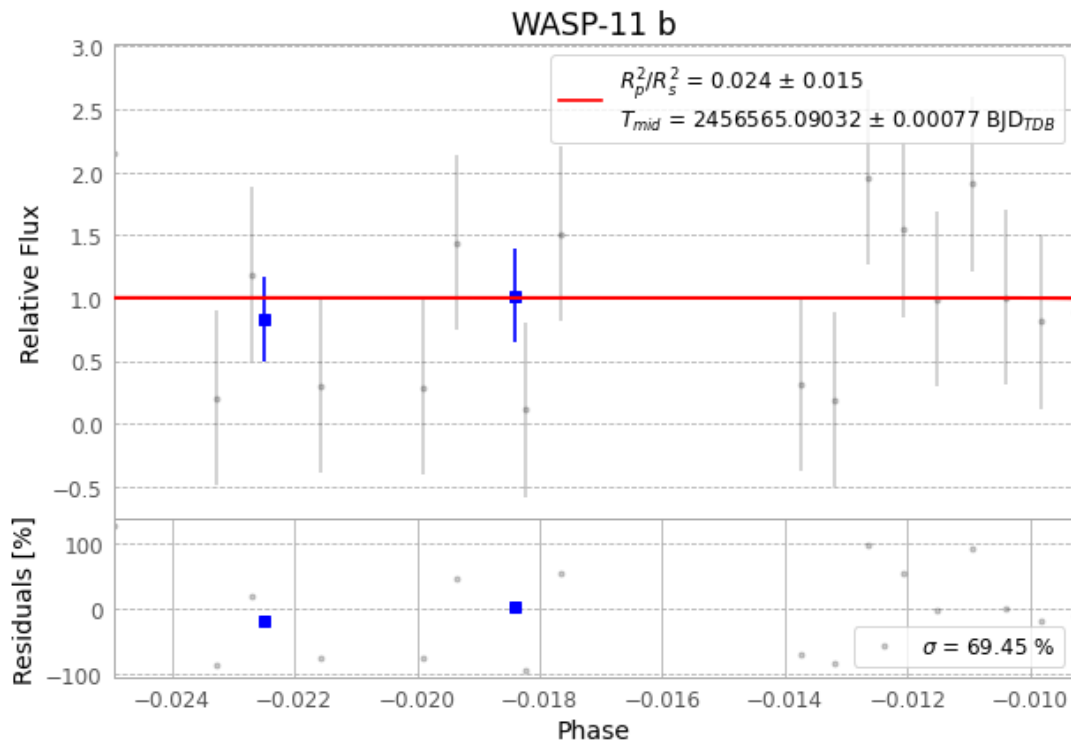


Figure 1 — FinalLightCurve_WASP-11 b_2013-10-02.png (SHA-256 484e1261135ee9d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [557, 389], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d47b321b41ffc7a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), HATP-10131003050912.FITS → HATP-10131003094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-131003.log (57e806270b37...), FinalLightCurve_WASP-11 b_2013-10-02.png (484e1261135e...), FinalLightCurve_WASP-11 b_2013-10-02.csv (23598985da73...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:45Z.